Spatial Stats - Homework 6.1.1

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1. I see an new benefit of reparametrization to ensure that our bayesian priors are orthogonal. In an arbitrary function with 3 or more parameters does these exist a mutually orthogonal set. I feel like this would be true, and it would require n invertable functions for n variates $f_{i_{i=1^n}}$ such that $\phi_i = f_i(\Theta)$ and $\text{Cov}(\phi_i, \phi_j) = 0$ and we can generate them by some PCA method or a "stochastic" gram-schmidt orthogonalization.

2. the notion of a process "rougher" than brownian motion is brought up, and I wanted to just confirm that this is in the path sense (how differentiable), not quadratic variation sense. I remember earlier in the year (on the snow day) the notion was brought up.

   Honestly the reason why I am getting confused is due to the fact that i know that for a R.V with finite second moment, the Q.V should be 0, but if we have a GP which is rougher (and by my assumption larger QV) than Brownian motion, it should have infinite second moment, but being a GP that is a contradiction.

3. The RMSE for $\hat{\phi}_2$ does not change much at all between models which was interesting. I feel like the reason might have to due with the magnitude of the η and β parameters as with smaller parameters for the eccentricity the less we need to adjust the estimates from a fixed model.